

# Transforming Unstructured Medical Data: A Digital System for Anonymized Electronic Health Records

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**Abstract**— The exponential growth of medical data presents significant challenges in the handling, processing, and secure storage of patient records. This paper introduces an advanced digital system designed to extract and store anonymized patient medical records within a national cancer research network. Utilizing modern web technologies and advanced Natural Language Processing techniques, the proposed solution effectively manages the complexities of unstructured and semi-structured data formats. The system, featuring a user-friendly web application, enhances data processing and accessibility. Validated through the successful extraction and structuring of data from over 40 patient records, it demonstrates significant potential in clinical research by reducing manual workload and improving data management. This paper details the system's architecture, functionalities, and benefits, emphasizing its impact on modernizing healthcare data management and its future applicability in the broader healthcare industry.

**Keywords**—electronic health records, natural language processing, healthcare data management

## I. INTRODUCTION

Electronic health records (EHR) have revolutionized the management of patient information, providing critical data for diagnostics and treatment. However, the exponential growth of medical records poses significant challenges in data handling, particularly concerning unstructured and semi-structured data formats. This research aims to develop a digital system that processes and stores medical records in an anonymized format, thereby supporting clinical research efforts.

The current context in healthcare highlights the importance of integrating technology into medical processes to improve the efficiency and reduce the workload for medical personnel. Digitalization offers significant benefits such as operational efficiency, accessibility, improved diagnosis and treatment, and better management of medical resources.

The digital transformation of healthcare is critical, considering the evolution of modern society, and many recent studies emphasize the importance of integrating modern technologies to enhance patient care and operational efficiency. The transition to electronic health records (EHRs) marked a significant leap forward in healthcare data management, but the real challenge lies in handling the vast amounts of unstructured and semi-structured data generated daily [1].

The challenge lies in translating traditional paper-based or unstructured medical records into a secure, scalable digital

format. This process involves overcoming issues like data integrity, and ensuring security and accessibility, considering the healthcare system that can be slow to adopt changes in procedures and modern solutions from the IT sector.

## II. RELATED WORK

The digitalization of patient medical records is a crucial aspect of modern healthcare, aiming to improve data accessibility, reduce errors, and enhance clinical research. Several solutions and technologies have been developed in this field, each addressing specific challenges related to the extraction, structuring, and storage of medical data. This section reviews notable contributions and existing technologies in the domain of electronic health records (EHR) and medical information extraction.

### A. Data Digitalization Projects in Healthcare

The field of healthcare data digitalization has seen substantial advancements driven by both scientific research and governmental initiatives. These projects aim to improve data extraction, processing, and utilization to enhance clinical research and patient care.

The WHO's Global Strategy on Digital Health 2020-2025 outlines a comprehensive approach to digital health, aiming to leverage technology to improve health outcomes worldwide. Similarly, the European Union's Digital Health Europe initiative focuses on facilitating cross-border health data exchange and supporting the development of digital health infrastructures [2].

Data-driven initiatives such as the DECOVRI Project, developed in response to the COVID-19 pandemic, focus on using Natural Language Processing (NLP) to extract relevant medical terms from clinical notes, particularly those related to COVID-19. This project demonstrated high accuracy in term extraction, crucial for effective treatment and research [3]. Initiatives launched by WHO in response to the COVID-19 pandemic have focused on digital tools to manage health crises effectively. These include mobile apps for health monitoring and guidelines for digital health implementation [4].

Various initiatives, such as those led by GE Healthcare and AstraZeneca, have integrated AI and big data into patient management and drug development. These projects leverage AI to enhance clinical decision-making and accelerate drug discovery, illustrating the transformative impact of digital technologies in healthcare [3]. The Robotics and Cloud-Assisted Healthcare System (ROCHAS) is another significant

initiative that combines robotics and cloud technologies to support healthcare for elderly populations. This project underscores the potential of integrating diverse technologies to enhance healthcare services [5].

There are significant governmental projects that focus on the digitalization of healthcare data, such as the Global Initiative on Digital Health (GIDH), managed by the World Health Organization (WHO). The project aims to accelerate progress towards health-related Sustainable Development Goals (SDGs) through digital health and includes various programs and tools to support digital health interventions globally. The European Healthcare Data Warehouse project focuses on creating interoperable data warehouse systems to support cross-border healthcare. The goal is to enhance data sharing and integration across different healthcare systems within Europe, facilitating better patient care and research [6].

The Health Information Technology for Economic and Clinical Health (HITECH) Act significantly boosted the adoption of EHRs in the United States. This legislation has led to nearly universal adoption of certified EHR systems among hospitals and a substantial increase in use among physicians, transforming clinical settings through digital technology [6].

### *B. Medical information extraction tools*

One of the pioneering systems for medical information extraction is MedLEE (Medical Language Extraction and Encoding System), developed at Columbia University. MedLEE uses natural language processing (NLP) techniques to extract and encode information from clinical texts, particularly radiology reports. It relies on a comprehensive knowledge base and a series of processing stages, including parsing, regularization, and encoding, to transform unstructured text into structured data [7].

The clinical Text Analysis and Knowledge Extraction System (cTAKES) is another significant open-source tool developed by the Mayo Clinic. Built on the Unstructured Information Management Architecture (UIMA) framework and utilizing OpenNLP, cTAKES is designed for processing clinical free-text and extracting relevant medical information. It includes components for sentence boundary detection, tokenization, part-of-speech tagging, and named entity recognition, making it a robust solution for clinical data extraction [8].

Google Cloud Healthcare API is a comprehensive platform that facilitates the storage, management, and analysis of healthcare data. It supports various healthcare data standards, such as HL7, FHIR, and DICOM, ensuring interoperability and compliance with industry regulations like HIPAA and GDPR. The platform also offers NLP capabilities to extract structured information from unstructured medical texts, enhancing the utility of EHR data in clinical research and patient care [9].

Amazon Comprehend Medical is a cloud-based NLP service that extracts medical information from unstructured text. It identifies medical conditions, medications, dosage, strengths, and frequencies, and also maps relationships between these entities. This service supports compliance with HIPAA and enables healthcare providers to process large volumes of medical data efficiently. Its ability to handle diverse data formats and provide scalable, secure data processing makes it a valuable tool for healthcare applications [10].

### *C. Databases for managing EHRs*

The growing volume and complexity of medical data have led to the adoption of NoSQL databases in healthcare. These databases offer flexible data models, horizontal scalability, and high availability, making them suitable for handling large-scale medical datasets. Studies have highlighted the benefits of NoSQL databases in managing EHRs, including their ability to store diverse data types (e.g., text, images, and multimedia) and support real-time data processing and analytics [11].

Ercan and Lane evaluated the performance of NoSQL databases in managing EHR systems, emphasizing their scalability and flexibility compared to traditional relational databases. Their findings suggest that NoSQL databases are better suited for handling the heterogeneous and dynamic nature of medical data, providing a more efficient solution for modern healthcare information systems [11].

### *D. Information Extraction Challenges*

Despite the advancements in medical information extraction, significant challenges remain. These include handling the variability and complexity of medical language, ensuring data privacy and security, and developing NLP models for languages other than English. The confidential nature of medical records further complicates the creation of training datasets for NLP models, limiting the applicability of existing solutions in non-English-speaking regions [12], [13], [14].

The digitalization and processing of electronic health records (EHRs) are crucial for modern healthcare systems, particularly for enhancing clinical research, improving patient care, and ensuring accurate diagnoses. This section explores various projects and technologies that have significantly contributed to this field, focusing on the methods and algorithms used for data extraction and processing in EHRs.

Enhancing EHR Usability with AI: Microsoft's acquisition of Nuance, a leader in AI and NLP technologies for healthcare, highlights the growing importance of integrating advanced NLP tools with EHR systems. Nuance's technologies improve the usability and accessibility of medical records by automatically extracting and categorizing information, aiding clinicians and researchers in their workflows [15].

Natural Language Processing (NLP) for EHRs: NLP has become an indispensable tool for extracting meaningful information from the unstructured text found in EHRs. Neural NLP models, including autoencoders, convolutional neural networks (CNNs), and recurrent neural networks (RNNs), have advanced the ability to process unstructured data in EHRs. These models effectively handle the variability and complexity of medical language. Sequence-to-sequence models, in particular, enable efficient text generation and classification, transforming raw clinical notes into structured data [15].

The Narrative Information Linear Extraction (NILE) project provides a high-performance NLP package for EHR analysis. NILE employs a modified prefix-tree search to efficiently parse and extract relevant data from clinical texts, emphasizing the importance of speed and accuracy in processing large volumes of unstructured medical data.

### III. PROPOSED SOLUTION

The digitalization and efficient management of patient medical records are critical for enhancing clinical research, improving patient care, and ensuring accurate and timely diagnoses. The increasing volume of unstructured and semi-structured medical data, however, presents significant challenges in terms of data extraction, structuring, and storage. Traditional methods of handling these records, often involving manual processing, are time-consuming and prone to errors.

To address these challenges, we propose a robust digital system designed specifically for structuring and storing anonymized patient medical records. This system leverages modern web technologies, advanced data processing algorithms, and a scalable NoSQL database to transform raw medical documents into structured, easily accessible data. The solution focuses on the following key objectives:

1. **Efficient Data Extraction:** Utilizing pattern recognition and NLP techniques to extract relevant information from medical records.
2. **Anonymization and Structuring:** Ensuring patient privacy by anonymizing sensitive information and organizing data into a structured format suitable for analysis and storage.
3. **Scalable and Secure Storage:** Implementing a NoSQL database to handle the large volumes of medical data with high availability and security.

The system's architecture depicted in Fig. 1 integrates a user-friendly web interface with a client-server architecture. The UI, developed using Angular, allows users to upload medical documents in .docx format, which are then processed by a Python server using the Flask framework. The extracted and anonymized data is stored in a Google Firestore NoSQL database, offering scalability and flexibility. The proposed system is composed of several integrated components:

- **User Interface (UI):** A web-based application built with Angular and Angular Material that allows users to upload medical documents, view processed data, and manage their records. The UI is designed to be intuitive and self-explanatory, ensuring minimal training time for medical staff.
- **Backend Server:** A Python server using the Flask framework to handle file uploads, process documents, and secure access to the database. The server employs various libraries for document parsing and data extraction.
- **NoSQL Database:** Google Firestore is utilized to store the structured medical data. This database is chosen for its flexibility, scalability, and ability to handle unstructured data types efficiently.

The integration of these components results in a streamlined workflow that significantly reduces the manual effort required to process medical records. By automating the extraction and structuring of data, the system enhances the accuracy and reliability of medical information available for clinical research.

The workflow is as follows: the user uploads a medical record; the system processes the record, extracts and anonymizes data; the user views and manages records.

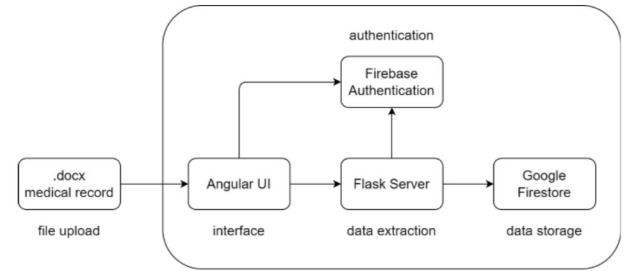


Fig. 1 System architecture

The key functionalities provisioned by the implemented system include:

- Uploading and processing medical records provided in .docx format.
- Extracting key medical information and storing structured data in a secure NoSQL database.
- Anonymizing patient data in compliance with GDPR regulations.
- Generating summaries of patient medical histories using NLP summarization techniques.
- Providing a user-friendly web interface for viewing and managing medical records.
- Supporting role-based access control for data security.

Furthermore, the implementation of robust security measures ensures that the system complies with industry standards and regulations such as GDPR, safeguarding patient privacy and maintaining the integrity of the data.

The proposed solution not only addresses the current challenges in medical data management but also lays the foundation for future enhancements, such as incorporating more advanced NLP techniques and expanding the system's capabilities to support a wider range of medical documents.

#### A. EHR Data Extraction and Pre-Processing

The system processes patient medical records from a semi-structured format, as illustrated by the sample records provided as Word (.docx) documents by the medical institutions involved in the CNCC project [16]. The relevance of the proposed methodology is supported by the requirements currently employed by the medical institutions to keep a physical or electronic document on record.

The key sections of the documents include: header information (medical center details, type of document, date and file number), patient identification (full name, national identification number, address, phone number, age, height, weight), diagnosis (detailed diagnosis of conditions), histopathological examinations (results of specific tests and biopsies), treatment information (details of previous treatments, interventions), clinical evolutions (observations and clinical notes), relevant tests and imaging (CT scans, MRI results, tumor markers), medications and dosages (details of administered medications and their dosages), discharge recommendations (post-treatment care instructions, scheduled follow-up appointments, emergency contact procedures for adverse effects). While each section is standardized to ensure consistency and ease of processing, there can be deviations

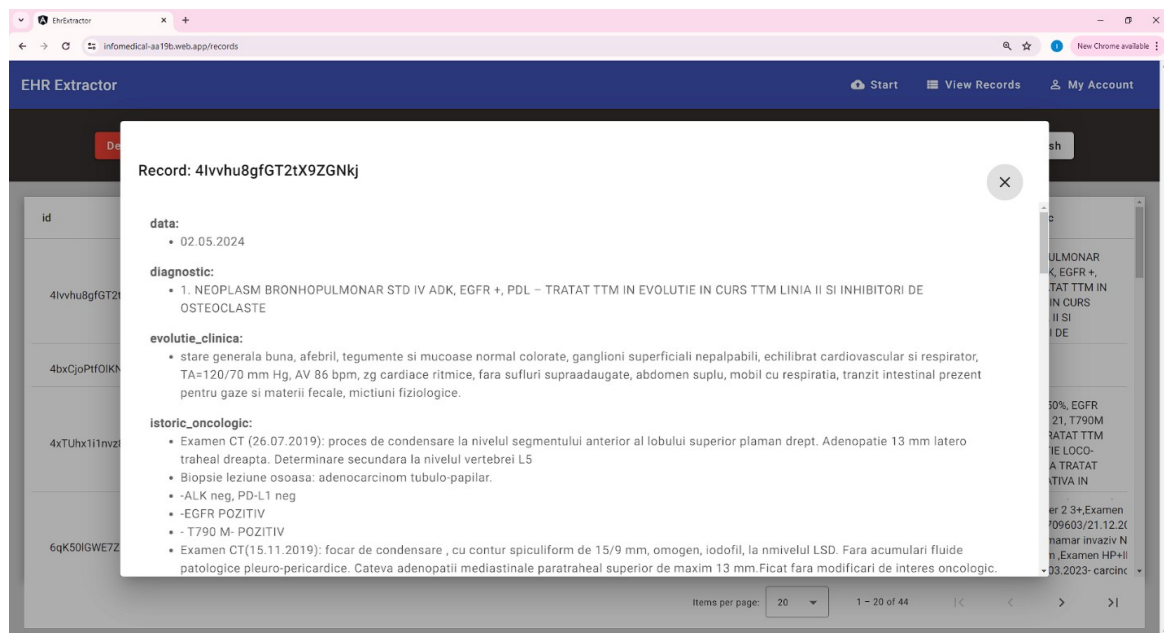


Fig. 2 Extracted patient data

from the template. From practice, it is revealed that the format of the documents changes over time, and some sections and contents are difficult to correlate in higher-level reports across different institutions.

By automating the extraction and structuring of these sections from medical records, the proposed system ensures that critical patient information is accurately captured and readily accessible, thereby supporting ongoing clinical research and improving patient outcomes. The system employs a component specifically designed to read and process the uploaded files, facilitating the extraction of both text and tables.

This preprocessing stage is crucial for handling the diverse formats and contents found in medical documents. To accurately identify and extract relevant sections from these documents, the system utilizes fuzzy matching algorithms. These algorithms employ a Levenshtein distance-based method to compare textual patterns and identify key information by comparing text against predefined patterns, allowing for the flexible recognition of section boundaries even when there are variations in the text.

In the extraction of specific document sections, the system uses fuzzy pattern matching to locate the start and end points of relevant sections. This method iterates through the document paragraphs, comparing text chunks to target patterns. When a match is found, the content between the matched patterns is extracted, ensuring that all relevant information is captured accurately.

By comparing the similarity between input text and predefined templates, the system can flexibly recognize section boundaries, even when there are variations in phrasing or format. This process allows for accurate extraction of critical data from diverse and non-standardized medical documents.

For extracting structured data from tables within the documents, the system implements specialized methods. These methods are tailored to handle different types of tables,

such as those containing general patient information, treatment details, and tumor markers. Each table extraction method processes rows and cells according to predefined configurations, mapping the extracted data to appropriate fields for structured storage.

The system employs a NoSQL database (Google Firestore) to handle the semi-structured nature of medical data. The database schema is designed to store records as collections of documents, with each document containing fields such as patient demographics, diagnoses, and treatment plans. This structure ensures flexibility in handling the varying data types and formats typical in medical records.

The backend server, built using the Flask framework, is responsible for handling file uploads and processing documents. It integrates with the document parsing components to extract necessary data, which is then stored in the NoSQL database. This integration ensures that the system can manage large volumes of heterogeneous data efficiently. Additionally, security and authentication are managed through Firebase, which restricts data access based on user roles and permissions, ensuring that sensitive medical information is protected.

The proposed solution effectively processes complex medical documents, extracting and structuring critical patient information for clinical research and improved healthcare delivery. This comprehensive approach not only enhances data accuracy and accessibility but also ensures compliance with data privacy regulations.

## B. Applications of NLP in EHR Data Processing

To demonstrate the applications of NLP, the primary goal was to develop a summarization system capable of condensing extensive medical records from the diagnostic section found in the provided EHR documents into concise summaries for each patient. The summarization process was implemented using the Hugging Face Transformers library, which provides state-of-the-art tools for NLP tasks [17]. This process involved several key NLP techniques:

- **Tokenization:** The text data was tokenized using the BERT tokenizer, which splits the text into tokens (words or subwords) suitable as input to the summarization model [18].
- **Text Chunking:** Due to model input length limitations, the text was chunked into smaller segments. Each segment was processed individually and later combined to create the final summary.
- **Summarization Model:** We employed the *facebook/bart-large-cnn* pre-trained transformer model known for its effectiveness in text summarization tasks. This model was chosen for its ability to generate coherent and contextually relevant summaries, with support for multiple languages [19].

The Natural Language Processing (NLP) component is central to the system's ability to handle unstructured medical records. We utilized the BERT model and the *facebook/bart-large-cnn* model for summarization and data extraction due to their state-of-the-art performance in understanding contextual language. BERT was chosen for its bidirectional training approach, allowing it to capture complex dependencies in medical texts. Additionally, *facebook/bart-large-cnn* was selected for its strong summarization capabilities. We made several custom configurations to better suit medical records, including fine-tuning on clinical datasets and adjusting hyperparameters to optimize performance on shorter text segments found in medical records.

#### IV. RESULTS

The implemented system successfully extracted key medical information from patient records and stored it securely in a NoSQL database. The user interface provides an intuitive platform for document upload and data visualization.

The input data consisted of anonymized medical records sourced from 40 patients within a national cancer research network. These records were processed to extract relevant fields using our custom NLP system. The data includes patient demographics, diagnosis details, and treatment history, structured in both free text and tabular formats. The testing environment consisted of a Python-based server using the Flask framework.

We evaluated the system on the following three metrics: precision, recall, and F1 score. These metrics measure the system's ability to accurately capture relevant information (precision), extract relevant tokens from the text (recall), and provide a balanced evaluation (F1 score).

In Fig. 2, the extracted data is shown in tabular format, with the general patient data (id, age, height, weight, address) provided as table columns, and the diagnostic data accessible by expanding a record. The now structured diagnostic data contains the following fields: the name of the diagnostic, clinical evolution, tumoral markers, recommendations at discharge, administered treatment.



Fig. 3 EHR adoption roadmap

The solution demonstrates significant potential in enhancing data management within medical research networks, and further applications in reducing the manual workload of managing local documents in favor of a cloud-based interface with big data capabilities.

The overall effectiveness of the proposed system can be benchmarked against industry standards such as the HIMSS Electronic Medical Record Adoption Model (EMRAM), which provides a framework to evaluate the degree of EHR adoption in healthcare settings [20]. The proposed application constitutes the transition from patient-centered reporting to remote data access, data integration and telemedicine (Fig. 3).

Preliminary results indicate that the proposed system aligns well with Stage 5 of the EMRAM, which involves improved data analysis capabilities and interoperability.

To demonstrate the higher-level potential of the proposed solution, the primary goal of NLP methods applied in this research was to develop a system that efficiently extracts patient medical records to facilitate better clinical decision-making and research. To assess the performance of the transformer-based summarization model, the BERTScore evaluation uses pre-trained BERT embeddings to evaluate the semantic similarity between the generated summaries and the original texts [21]. It provides precision, recall, and F1 score metrics, offering a comprehensive measure of summary quality. The BERTScore evaluation results for our summarization model are as follows:

- **Precision:** A precision score of 0.7385 indicates that approximately 73.85% of the tokens in the generated summaries are relevant and present in the original texts. This suggests that the model effectively captures a significant portion of relevant information from the original texts.
- **Recall:** A recall score of 0.7055 shows that about 70.55% of the relevant tokens in the original texts are included in the generated summaries. This demonstrates the model's ability to extract a substantial amount of relevant information.
- **F1 Score:** The F1 Score of 0.7200, which is the harmonic mean of precision and recall, reflects a balanced performance of the summarization model. It indicates that, on average, 72.00% of the tokens are correctly summarized.

Therefore, by leveraging advanced NLP techniques and evaluation metrics, the proposed system demonstrates significant potential to enhance healthcare delivery, streamline research processes, and improve patient outcomes.



## V. CONCLUSION

This paper highlights the motivation, architecture, functionalities, and benefits of the proposed digital system for medical data management. By anonymizing and structuring medical data, the system supports clinical research and data analysis, and provides tools for generating comprehensive reports and visualizing patient data trends. The web-based solution facilitates quick access to patient data for healthcare professionals, enabling remote access to records, supporting telemedicine and modern healthcare practices. Benefits include operational efficiency, by reducing the resources required for managing medical information and minimizing manual effort needed to work with local documents.

The proposed solution facilitates the applications of NLP in processing patient data, intended for various practical applications within the healthcare domain, from clinical decision support, where summarized records provide clinicians with quick access to essential patient information, facilitating more efficient and informed decision-making, to research and analysis, where researchers can utilize summarized data to conduct large-scale studies and identify trends or patterns in patient populations. Finally, to improve communication with patients, summarized reports can be used to convey complex medical information to patients in a more understandable manner, enhancing patient engagement and compliance with treatment plans.

Further developments include more advanced NLP techniques and broader support for various types of medical documents, emphasizing ongoing improvement in data processing and system scalability to handle increasing volumes of medical data.

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